



VIGNAN'S

FOUNDATION FOR SCIENCE, TECHNOLOGY & RESEARCH

(Deemed to be University) - Estd. u/s 3 of UGC Act 1956

```
test> use salesDB
switched to db salesDB
```

```
salesDB> db.sales.insertMany([
...   { productId: 1, productName: "Laptop", quantity: 5, price: 1000, date: "2024-09-20", city: "New York" },
...   { productId: 2, productName: "Phone", quantity: 3, price: 600, date: "2024-09-21", city: "Los Angeles" },
...   { productId: 1, productName: "Laptop", quantity: 2, price: 1000, date: "2024-09-22", city: "Chicago" },
...   { productId: 3, productName: "Tablet", quantity: 4, price: 400, date: "2024-09-20", city: "New York" },
...   { productId: 4, productName: "Chair", quantity: 1, price: 100, date: "2024-09-23", city: "Houston" },
...   { productId: 5, productName: "Desk", quantity: 6, price: 300, date: "2024-09-24", city: "New York" },
...   { productId: 3, productName: "Tablet", quantity: 3, price: 400, date: "2024-09-25", city: "Los Angeles" },
...   { productId: 2, productName: "Phone", quantity: 5, price: 600, date: "2024-09-26", city: "Chicago" }
... ]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('66fa3c2405742e43a92710bc'),
    '1': ObjectId('66fa3c2405742e43a92710bd'),
    '2': ObjectId('66fa3c2405742e43a92710be'),
    '3': ObjectId('66fa3c2405742e43a92710bf'),
    '4': ObjectId('66fa3c2405742e43a92710c0'),
    '5': ObjectId('66fa3c2405742e43a92710c1'),
    '6': ObjectId('66fa3c2405742e43a92710c2'),
    '7': ObjectId('66fa3c2405742e43a92710c3')
  }
}
salesDB>
```

Compass

127.0.0.1:27017 > salesDB > sales

Documents 4 Aggregations Schema Indexes 1 Validation

Search connections

127.0.0.1:27017

- admin
- college
- config
- local
- salesDB
 - sales
- school
- vignan
- localhost:27017

Type a query: (field: 'value') or [Generate query](#)

Explain Reset Find Options

ADD DATA EXPORT DATA UPDATE DELETE

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```
{ "_id": ObjectId('66fa3c2405742e43a92710bc'),
  productId: 1,
  productName: "Laptop",
  quantity: 5,
  price: 1000,
  date: "2024-09-20",
  city: "New York" }

{ "_id": ObjectId('66fa3c2405742e43a92710bd'),
  productId: 2,
  productName: "Phone",
  quantity: 3,
  price: 600,
  date: "2024-09-21",
  city: "Los Angeles" }

{ "_id": ObjectId('66fa3c2405742e43a92710be'),
  productId: 1,
  productName: "Laptop",
  quantity: 2,
  price: 1000,
  date: "2024-09-22",
  city: "Chicago" }

{ "_id": ObjectId('66fa3c2405742e43a92710bf'),
  productId: 3,
  productName: "Tablet",
  quantity: 4,
  price: 400,
  date: "2024-09-20",
  city: "New York" }
```

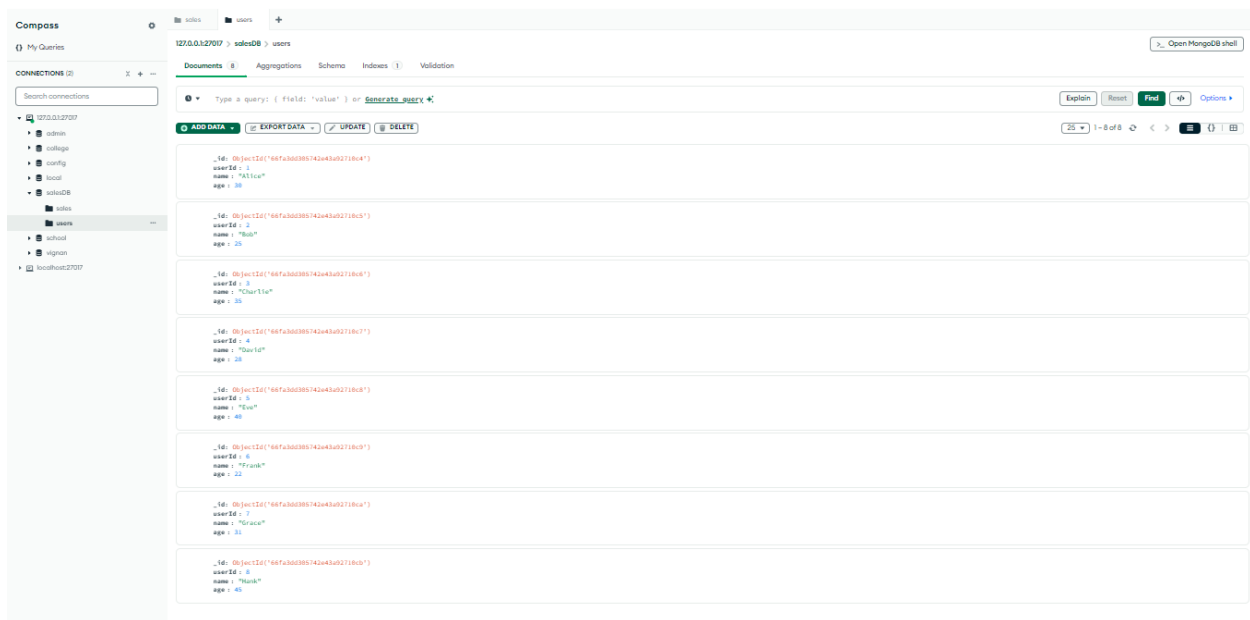
```
._id: ObjectId('66fa3c2485742e43a92710c9')
productid : 4
productName : "Chair"
quantity : 1
price : 100
date : "2024-09-23"
city : "Houston"
```

```
._id: ObjectId('66fa3c2485742e43a92710c1')
productid : 5
productName : "Desk"
quantity : 6
price : 300
date : "2024-09-24"
city : "New York"
```

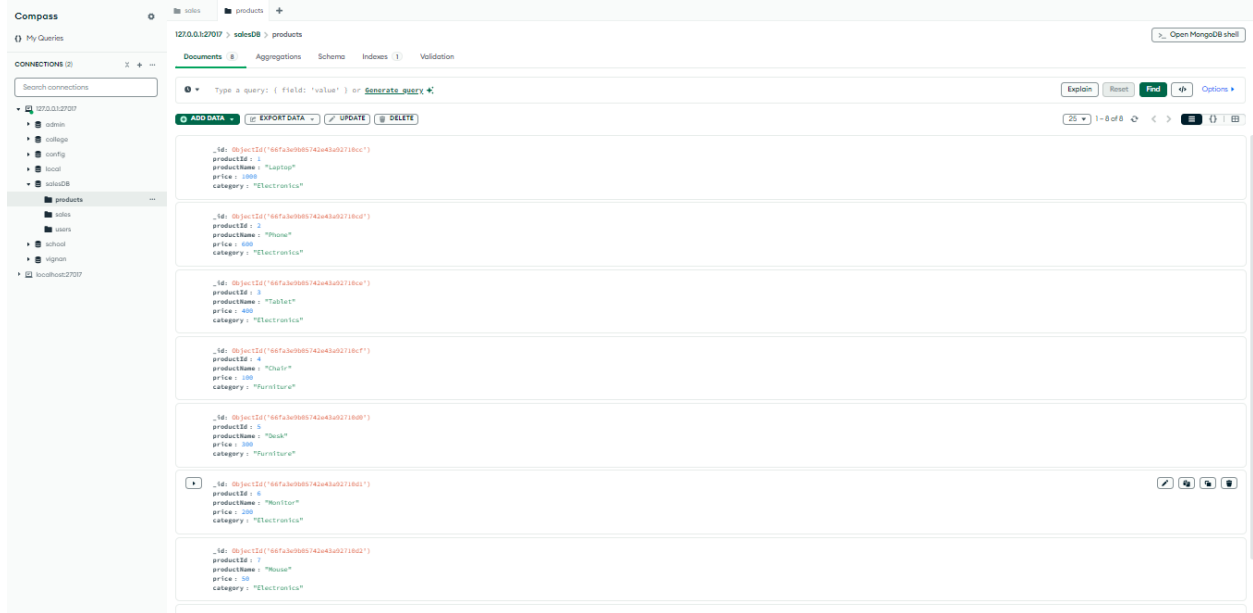
```
._id: ObjectId('66fa3c2485742e43a92710c2')
productid : 3
productName : "Tablet"
quantity : 3
price : 400
date : "2024-09-25"
city : "Los Angeles"
```

```
._id: ObjectId('66fa3c2485742e43a92710c3')
productid : 2
productName : "Phone"
quantity : 5
price : 600
date : "2024-09-26"
city : "Chicago"
```

```
salesDB> db.users.insertMany([
...   { userId: 1, name: "Alice", age: 30 },
...   { userId: 2, name: "Bob", age: 25 },
...   { userId: 3, name: "Charlie", age: 35 },
...   { userId: 4, name: "David", age: 28 },
...   { userId: 5, name: "Eve", age: 40 },
...   { userId: 6, name: "Frank", age: 22 },
...   { userId: 7, name: "Grace", age: 31 },
...   { userId: 8, name: "Hank", age: 45 }
... ]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('66fa3dd305742e43a92710c4'),
    '1': ObjectId('66fa3dd305742e43a92710c5'),
    '2': ObjectId('66fa3dd305742e43a92710c6'),
    '3': ObjectId('66fa3dd305742e43a92710c7'),
    '4': ObjectId('66fa3dd305742e43a92710c8'),
    '5': ObjectId('66fa3dd305742e43a92710c9'),
    '6': ObjectId('66fa3dd305742e43a92710ca'),
    '7': ObjectId('66fa3dd305742e43a92710cb')
  }
}
salesDB>
```



```
salesDB> db.products.insertMany([
...   { productId: 1, productName: "Laptop", price: 1000, category: "Electronics" },
...   { productId: 2, productName: "Phone", price: 600, category: "Electronics" },
...   { productId: 3, productName: "Tablet", price: 400, category: "Electronics" },
...   { productId: 4, productName: "Chair", price: 100, category: "Furniture" },
...   { productId: 5, productName: "Desk", price: 300, category: "Furniture" },
...   { productId: 6, productName: "Monitor", price: 200, category: "Electronics" },
...   { productId: 7, productName: "Mouse", price: 50, category: "Electronics" },
...   { productId: 8, productName: "Keyboard", price: 80, category: "Electronics" }
... ]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('66fa3e9b05742e43a92710cc'),
    '1': ObjectId('66fa3e9b05742e43a92710cd'),
    '2': ObjectId('66fa3e9b05742e43a92710ce'),
    '3': ObjectId('66fa3e9b05742e43a92710cf'),
    '4': ObjectId('66fa3e9b05742e43a92710d0'),
    '5': ObjectId('66fa3e9b05742e43a92710d1'),
    '6': ObjectId('66fa3e9b05742e43a92710d2'),
    '7': ObjectId('66fa3e9b05742e43a92710d3')
  }
}
salesDB>
```



```
_id: ObjectId('66fa3e9b85742e43a92718d3')
productId: 8
productName: "Keyboard"
price: 80
category: "Electronics"
```

```
salesDB> db.blogPosts.insertMany([
...   { postId: 1, title: "Tech Trends", tags: ["technology", "innovation"] },
...   { postId: 2, title: "Interior Design", tags: ["design", "furniture"] },
...   { postId: 3, title: "Travel Tips", tags: ["travel", "adventure"] },
...   { postId: 4, title: "Fitness Advice", tags: ["health", "fitness"] },
...   { postId: 5, title: "Programming 101", tags: ["coding", "technology"] },
...   { postId: 6, title: "Healthy Eating", tags: ["food", "health"] },
...   { postId: 7, title: "Remote Work", tags: ["work", "productivity"] },
...   { postId: 8, title: "Investing Basics", tags: ["finance", "investing"] }
... ]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('66fa404305742e43a92710e4'),
    '1': ObjectId('66fa404305742e43a92710e5'),
    '2': ObjectId('66fa404305742e43a92710e6'),
    '3': ObjectId('66fa404305742e43a92710e7'),
    '4': ObjectId('66fa404305742e43a92710e8'),
    '5': ObjectId('66fa404305742e43a92710e9'),
    '6': ObjectId('66fa404305742e43a92710ea'),
    '7': ObjectId('66fa404305742e43a92710eb')
  }
}
salesDB>
```

CONNECTIONS (2) ✕ + ...

Search connections

- 127.0.0.1:27017
 - admin
 - collage
 - config
 - local
 - salesDB
 - blogPosts
 - products
 - sales
 - users
 - school
 - vignan
- localhost:27017

Documents | 8 Aggregations Schema Indexes (1) Validation

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA EXPORT DATA UPDATE DELETE

```

{
  "_id": ObjectId("66fa404385742e43a92718e4"),
  "postId": 1,
  "title": "Tech Trends",
  "tags": Array (2)
}

{
  "_id": ObjectId("66fa404385742e43a92718e5"),
  "postId": 2,
  "title": "Interior Design",
  "tags": Array (2)
}

{
  "_id": ObjectId("66fa404385742e43a92718e6"),
  "postId": 3,
  "title": "Travel Tips",
  "tags": Array (2)
}

{
  "_id": ObjectId("66fa404385742e43a92718e7"),
  "postId": 4,
  "title": "Fitness Advice",
  "tags": Array (2)
}

{
  "_id": ObjectId("66fa404385742e43a92718e8"),
  "postId": 5,
  "title": "Programming 101",
  "tags": Array (2)
}

{
  "_id": ObjectId("66fa404385742e43a92718e9"),
  "postId": 6,
  "title": "Healthy Eating",
  "tags": Array (2)
}

{
  "_id": ObjectId("66fa404385742e43a92718ea"),
  "postId": 7,
  "title": "Remote Work",
  "tags": Array (2)
}

{
  "_id": ObjectId("66fa404385742e43a92718eb"),
  "postId": 8,
  "title": "Investing Basics",
  "tags": Array (2)
}

```

```

salesDB> db.reviews.insertMany([
...   { productId: 1, rating: 5 },
...   { productId: 2, rating: 4 },
...   { productId: 1, rating: 3 },
...   { productId: 3, rating: 4 },
...   { productId: 4, rating: 5 },
...   { productId: 5, rating: 2 },
...   { productId: 6, rating: 5 },
...   { productId: 7, rating: 4 }
... ]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('66fa41a305742e43a92710ec'),
    '1': ObjectId('66fa41a305742e43a92710ed'),
    '2': ObjectId('66fa41a305742e43a92710ee'),
    '3': ObjectId('66fa41a305742e43a92710ef'),
    '4': ObjectId('66fa41a305742e43a92710f0'),
    '5': ObjectId('66fa41a305742e43a92710f1'),
    '6': ObjectId('66fa41a305742e43a92710f2'),
    '7': ObjectId('66fa41a305742e43a92710f3')
  }
}
salesDB>

```

127.0.0.1:27017

admin

college

config

local

salesDB

- blogPosts
- products
- reviews
- sales
- users

school

vignan

localhost:27017

ADD DATA

EXPORT DATA

UPDATE

DELETE

_id: ObjectId('66fa41a385742e43a92710ec')

productId: 1

rating: 5

_id: ObjectId('66fa41a385742e43a92710ed')

productId: 2

rating: 4

_id: ObjectId('66fa41a385742e43a92710ee')

productId: 1

rating: 3

_id: ObjectId('66fa41a385742e43a92710ef')

productId: 3

rating: 4

_id: ObjectId('66fa41a385742e43a92710f0')

productId: 4

rating: 5

_id: ObjectId('66fa41a385742e43a92710f1')

productId: 5

rating: 2

_id: ObjectId('66fa41a385742e43a92710f2')

productId: 6

rating: 5

_id: ObjectId('66fa41a385742e43a92710f3')

productId: 7

rating: 4

```
salesDB> db.orders.insertMany([
...   { orderId: 1, customerId: 1, totalAmount: 200 },
...   { orderId: 2, customerId: 2, totalAmount: 150 },
...   { orderId: 3, customerId: 1, totalAmount: 300 },
...   { orderId: 4, customerId: 3, totalAmount: 500 },
...   { orderId: 5, customerId: 4, totalAmount: 100 },
...   { orderId: 6, customerId: 5, totalAmount: 250 },
...   { orderId: 7, customerId: 6, totalAmount: 300 },
...   { orderId: 8, customerId: 1, totalAmount: 600 }
... ]);
{   '4': ObjectId('66fa3f6c05742e43a92710d8'),
    acknowledged: true,fa3f6c05742e43a92710d9'),
    insertedIds: {d('66fa3f6c05742e43a92710da'),
                  '0': ObjectId('66fa41ef05742e43a92710f4'),
                  '1': ObjectId('66fa41ef05742e43a92710f5'),
                  '2': ObjectId('66fa41ef05742e43a92710f6'),
                  '3': ObjectId('66fa41ef05742e43a92710f7'),
                  '4': ObjectId('66fa41ef05742e43a92710f8'),
                  '5': ObjectId('66fa41ef05742e43a92710f9'),
                  '6': ObjectId('66fa41ef05742e43a92710fa'),
                  '7': ObjectId('66fa41ef05742e43a92710fb')
    }
}
salesDB> ■
```

CONNECTIONS (2) X + ...

Search connections

- 127.0.0.1:27017
 - admin
 - college
 - config
 - local
 - salesDB
 - blogPosts
 - orders
 - products
 - reviews
 - sales
 - users
 - school
 - vignan
- localhost:27017

Documents (8) Aggregations Schema Indexes (1) Validation

Type a query: { field: 'value' } or [Generate query](#) +

ADD DATA EXPORT DATA UPDATE DELETE

```
{ "_id": ObjectId("66fa41ef85742e43a92718f4"), "orderId": 1, "customerId": 1, "totalAmount": 200 }
```

```
{ "_id": ObjectId("66fa41ef85742e43a92718f5"), "orderId": 2, "customerId": 2, "totalAmount": 150 }
```

```
{ "_id": ObjectId("66fa41ef85742e43a92718f6"), "orderId": 3, "customerId": 1, "totalAmount": 300 }
```

```
{ "_id": ObjectId("66fa41ef85742e43a92718f7"), "orderId": 4, "customerId": 3, "totalAmount": 500 }
```

```
{ "_id": ObjectId("66fa41ef85742e43a92718f8"), "orderId": 5, "customerId": 4, "totalAmount": 100 }
```

```
{ "_id": ObjectId("66fa41ef85742e43a92718f9"), "orderId": 6, "customerId": 5, "totalAmount": 250 }
```

```
{ "_id": ObjectId("66fa41ef85742e43a92718fa"), "orderId": 7, "customerId": 6, "totalAmount": 300 }
```

```
{ "_id": ObjectId("66fa41ef85742e43a92718fb"), "orderId": 8, "customerId": 1, "totalAmount": 600 }
```

Question 1: Write an aggregation query to find the total sales amount for each product in the sales collection.

Answer:


```

salesDB> db.sales.aggregate([
...   {
...     $group: {
...       _id: "$productName",
...       totalSalesAmount: { $sum: { $multiply: ["$quantity", "$price"] } }
...     }
...   }
... ]);
[
  { _id: 'Phone', totalSalesAmount: 4800 },
  { _id: 'Desk', totalSalesAmount: 1800 },
  { _id: 'Laptop', totalSalesAmount: 7000 },
  { _id: 'Tablet', totalSalesAmount: 2800 },
  { _id: 'Chair', totalSalesAmount: 100 }
]
salesDB>

```

Question 2: Write an aggregation query to calculate the average age of users in the users collection.

Answer:

```

salesDB> db.users.aggregate([
...   {
...     $group: {
...       _id: null,
...       averageAge: { $avg: "$age" }
...     }
...   }
... ]);
[ { _id: null, averageAge: 32 } ]
salesDB>

```

Question 3: Write an aggregation query to find the maximum and minimum price of products in the products collection.

Answer:

```

salesDB> db.products.aggregate([
...   {
...     $group: {
...       _id: null,
...       maxPrice: { $max: "$price" },
...       minPrice: { $min: "$price" }
...     }
...   }
... ]);
[ { _id: null, maxPrice: 1000, minPrice: 50 } ]
salesDB>

```

Question 4: Write an aggregation query to count the number of products in each category in the products collection.

Answer:

```

salesDB> db.products.aggregate([
...   {
...     $group: {
...       _id: "$category",
...       productCount: { $sum: 1 }
...     }
...   }
... ]);
[
  { _id: 'Electronics', productCount: 6 },
  { _id: 'Furniture', productCount: 2 }
]
salesDB>

```

Question 5: Write an aggregation query to list all unique tags used in the blogPosts collection.

Answer:

```

salesDB> db.blogPosts.aggregate([
...   { $unwind: "$tags" },
...   {
...     $group: {
...       _id: "$tags"
...     }
...   },
...   { $sort: { _id: 1 } }
... ]);
[
  { _id: 'adventure' },
  { _id: 'coding' },
  { _id: 'design' },
  { _id: 'finance' },
  { _id: 'fitness' },
  { _id: 'food' },
  { _id: 'furniture' },
  { _id: 'health' },
  { _id: 'innovation' },
  { _id: 'investing' },
  { _id: 'productivity' },
  { _id: 'technology' },
  { _id: 'travel' },
  { _id: 'work' }
]
salesDB>

```

Question 6: Write an aggregation query to find the total quantity sold per day from the sales collection.

Answer:

```

salesDB> db.sales.aggregate([
...   {
...     $group: {
...       _id: "$date",
...       totalQuantity: { $sum: "$quantity" }
...     }
...   }
... ]);
[
  { _id: '2024-09-21', totalQuantity: 3 },
  { _id: '2024-09-25', totalQuantity: 3 },
  { _id: '2024-09-26', totalQuantity: 5 },
  { _id: '2024-09-22', totalQuantity: 2 },
  { _id: '2024-09-24', totalQuantity: 6 },
  { _id: '2024-09-23', totalQuantity: 1 },
  { _id: '2024-09-20', totalQuantity: 9 }
]
salesDB>

```

Question 7: Write an aggregation query to calculate the total revenue generated from each city in the sales collection.

Answer:

```

salesDB> db.sales.aggregate([
...   {
...     $group: {
...       _id: "$city",
...       totalRevenue: { $sum: { $multiply: ["$quantity", "$price"] } }
...     }
...   }
... ]);
[
  { _id: 'New York', totalRevenue: 8400 },
  { _id: 'Houston', totalRevenue: 100 },
  { _id: 'Chicago', totalRevenue: 5000 },
  { _id: 'Los Angeles', totalRevenue: 3000 }
]
salesDB>

```

Question 8: Write an aggregation query to find the total number of orders placed by each customer in the orders collection.

Answer:

```

salesDB> db.orders.aggregate([
...   {
...     $group: {
...       _id: "$customerId",
...       totalOrders: { $sum: 1 }
...     }
...   }
... ]);
[
  { _id: 3, totalOrders: 1 },
  { _id: 1, totalOrders: 3 },
  { _id: 5, totalOrders: 1 },
  { _id: 2, totalOrders: 1 },
  { _id: 4, totalOrders: 1 },
  { _id: 6, totalOrders: 1 }
]
salesDB>

```

Question 9: Write an aggregation query to find the average rating for each product in the reviews collection.

Answer:

```

salesDB> db.reviews.aggregate([
...   {
...     $group: {
...       _id: "$productId",
...       averageRating: { $avg: "$rating" }
...     }
...   }
... ]);
[
  { _id: 1, averageRating: 4 },
  { _id: 3, averageRating: 4 },
  { _id: 5, averageRating: 2 },
  { _id: 2, averageRating: 4 },
  { _id: 4, averageRating: 5 },
  { _id: 7, averageRating: 4 },
  { _id: 6, averageRating: 5 }
]
salesDB>

```

Question 10: Write an aggregation query to find the most popular category based on the total number of products sold in the sales collection.

Answer:

```
salesDB> db.sales.aggregate([
...   {
...     $lookup: {
...       from: "products",
...       localField: "productId",
...       foreignField: "productId",
...       as: "productDetails"
...     }
...   },
...   { $unwind: "$productDetails" },
...   {
...     $group: {
...       _id: "$productDetails.category",
...       totalQuantitySold: { $sum: "$quantity" }
...     }
...   },
...   { $sort: { totalQuantitySold: -1 } },
...   { $limit: 1 }
... ]);
[ { _id: 'Electronics', totalQuantitySold: 22 } ]
salesDB> ■
```